

Chapter 20 — Complications of Urologic Surgery

- **Minimally invasive surgery (MIS) lowered morbidity but introduced new complications.** Index oncological and reconstructive operations are now largely laparoscopic or robotic; trainee exposure to open surgery has fallen, but the principles of managing major complications are unchanged.
- **Learning curves have shortened** (open radical prostatectomy 100–1000 cases; robotic about 40), yet **complication rates have not fallen in step**, partly because many young surgeons are still on the curve.
- **Laparoscopic risk factors:** complex procedures (partial nephrectomy, cystectomy), **operating time >4 h**, **serum creatinine >1.5 mg/dL**.
- **Complexity versus experience:** complications rise up to **ninefold with complexity** and fall **fourfold with experience**. Yet absolute numbers can **rise with experience**, because harder cases are taken on; complication and conversion rates climb from easy to very difficult operations.
- **Laparoscopy carries every open complication plus its own** (pneumoperitoneum, access), so a **high index of suspicion** is needed during and after surgery.
- **Technology also treats complications** (interventional radiology, MIS repair). Recent graduates may meet complications never seen in training, with no specific guidelines; **simulators and animal models** may bridge the gap.
- **Scope:** positioning, access, vascular, urinary, bowel and visceral complications; procedure-specific endourological complications are covered in their own chapters.

Concept — *complication burden reflects complexity multiplied by inexperience: experience lowers risk per case but moves surgeons to harder cases, so absolute numbers need not fall.*

CLASSIFICATION OF COMPLICATIONS

The Clavien-Dindo Scale

- **The recommended uniform reporting system.** It grades by **invasiveness of the treatment needed**, not by severity of the event.
- Not designed for urology but **validated as simple, reproducible and applicable**; urological use roughly doubled between 2010 and 2012.
- **Limitations:** management-based rather than morbidity-based; **no early/late distinction**; **omits intraoperative events**; may be **unsuitable in paediatric urology**.
- **High interobserver agreement**, even in ambiguous cases; about 80% trainee–attending concordance. The most widely used tool.
- **Percutaneous nephrolithotomy (PCNL):** good concordance for **higher grades**, **poor at grade 3 or below**, because perceived risk and required management diverge. Since applied to radical nephrectomy and robotic prostatectomy.
- **Standardisation is the cornerstone of error reduction**; future procedural guidelines may standardise steps and define surgical errors.

TABLE 20.1 THE CLAVIEN-DINDO CLASSIFICATION

Grade	Definition	Includes / key detail
I	Deviation from normal course without drugs or surgical, endoscopic or radiological intervention	Antiemetics, antipyretics, analgesics, diuretics, electrolytes, physiotherapy; wound opened at the bedside

Grade	Definition	Includes / key detail
II	Drugs beyond those allowed in grade I	Blood transfusion; total parenteral nutrition (TPN)
IIIa / IIIb	Surgical, endoscopic or radiological intervention	IIIa without , IIIb under general anaesthesia
IVa / IVb	Life-threatening; intermediate or intensive care (IC/ICU)	IVa single organ (includes dialysis) ; IVb multiorgan. CNS events count (haemorrhage, ischaemic stroke, subarachnoid bleed), not transient ischaemic attack (TIA)
V	Death	—
Suffix “d”	Complication present at discharge (“disability”)	Added to the grade; flags need for follow-up

Treatment modality is the cornerstone. Seven grades; subdivisions allow rare, severe complications to be pooled.

- **Pattern:** I–II drugs; III intervention, split by anaesthesia; IV organ failure, split by number of organs; V death.
- **Traps:** transfusion is **II**; bedside wound opening is **I**; dialysis is **IVa**; TIA excluded; “d” is a suffix, not a grade; III split by anaesthesia, not procedure size.

Concept — *Clavien-Dindo grades the response, not the event: the same complication is grade I or IIIb purely according to what treated it.*

PHYSIOLOGIC COMPLICATIONS

- **Pneumoperitoneum:** bradycardia from peritoneal gas distension; **hypercarbia and acidosis** from carbon dioxide (CO₂); **reduced respiratory reserve with raised airway pressure** (detailed in the laparoscopy chapters).

POSITIONAL AND NEUROMUSCULAR COMPLICATIONS

- **Mechanism:** direct surgical trauma or positioning, mostly a nerve **compressed against the patient's body or the table**.
- **Positions:** supine, lateral decubitus (flank), lithotomy, and robotic **steep Trendelenburg**, which compromises all four limbs.
- **Obesity raises risk twice over:** weight on pressure points and longer operations.
- **Incidence** about 1% (large laparoscopic series), **<3%** in urology generally, up to about 6% in robotic series. **Permanent deficit is very rare;** most recover spontaneously.

Positional Nerve Injury

- **Prolonged compression**, with or without padding; presents **immediately after surgery** with sensory loss, paraesthesia or weakness in the nerve's territory.

POSITIONAL NERVE INJURIES BY POSITION

Position	Nerve	Mechanism	Deficit
Lithotomy	Posterior tibial	Back of knee against stirrup	Weak plantarflexion ; numb sole and lateral foot; calf paraesthesia
Lithotomy	Common peroneal	Stirrup at the fibular neck	Foot drop ; weak eversion
Lithotomy	Pudendal	Leg traction; stirrup compression	Perineal numbness; rarely incontinence
Exaggerated lithotomy	Obturator	Hip hyperflexion	Weak adduction

Position	Nerve	Mechanism	Deficit
Lateral	Peroneal (dependent leg)	Unpadded leg against table	Ipsilateral foot drop
Lateral	Brachial plexus	Dependent shoulder or arm under rib cage	Shoulder pain, arm weakness; axillary roll (flank, prone)
Supine or lateral	Brachial plexus	Most cases: abduction >90° , external rotation, posterior shoulder displacement	Shoulder pain, arm weakness; early physiotherapy
Supine	Sciatic	Inadequate padding	Numb lateral calf and foot
Supine, arm off board	Radial / median	Supination → radial; pronation → median	Radial: wrist drop . Median: weak grip, palmar numbness
Supine; arms tucked (robotic)	Ulnar	Forearm hyperextended and pronated; elbow flexed 90° across chest; pressure against torso or table	Weak grip or claw hand (MCP extended, IP flexed, 4th–5th fingers)
Prone (PCNL)	Ilioinguinal	Lower bolster too firm	Sensory neuropathy
Prone; reverse Trendelenburg	Anterior tibial	Foot held in plantarflexion (weight on unsupported soles)	Foot drop
Prone	Lateral femoral cutaneous	Lateral thigh pressure	Numb lateral and anterior thigh

- **Pattern:** the injured nerve is the one crossing a hard edge in that position.
- **Traps:** radial versus median turns on **supination versus pronation**; foot drop may be peroneal (lithotomy, lateral) or anterior tibial (prone). **Check arm abduction and fixation before draping**, since it is invisible afterwards. With robotic arms tucked, pad the hands and keep the wrists straight.

Operative Nerve Injury

- The nerves most often met are the **genitofemoral, ilioinguinal and obturator**.

OPERATIVE NERVE INJURIES

Nerve	Operation	Mechanism / deficit	Prevention / repair
Genitofemoral (also femoral, lateral cutaneous)	Reimplantation, psoas hitch (neuropathy about 5%)	Entrapped by psoas anchoring suture → scrotal and medial thigh numbness; paraesthesia to severe pain = “ psoas syndrome ”	Sutures longitudinal, along the tendon , not across it
Femoral	Psoas hitch	In very thin patients, nerve mistaken for an indistinct tendon	Positively identify the tendon
Ilioinguinal	Inguinal orchidectomy; hernia repair (most injuries)	Inguinal and lateral hemiscrotal pain, paraesthesia	—
Obturator	Pelvic lymph node dissection; robotic cystectomy (lateral landmark)	Clipped or cut → medial thigh numbness, adductor weakness . Proximal segment near the internal iliac vein most at risk (about 78%)	Clip only with the nerve in view; remove clips on it . Transection → tension-free 7-0 non-absorbable repair (open or robotic) plus early intensive physiotherapy. Rates fell with experience (about 5% → 0.1%)

Nerve	Operation	Mechanism / deficit	Prevention / repair
Intercostal / subcostal (T10–T12)	Open flank incision, lumbotomy	Run along the inferior rib border with the vessels	Incise from the upper border of the 11th rib or tip of the 12th towards the umbilicus; layered closure (bulky single layer entraps)

- **Obturator exposure:** displace the external iliac vessels and nodal packets in the **triangle of Marcille** medially.

Gluteal Compartment Syndrome and Lower Extremity Compartment Syndrome

GLUTEAL VERSUS LOWER EXTREMITY COMPARTMENT SYNDROME

	Gluteal	Lower extremity
Frequency	Very rare ; usually trauma or medical; positional cases in obese patients after long operations	1 in 3500 overall; up to 1 in 500 radical cystectomies
Setting	Obesity, prolonged surgery	Lithotomy , especially early MIS experience (long cases)
Mechanism	Compression	Low limb perfusion pressure → ischaemia → rising compartment pressure
Signs	Gluteal tenderness; reduced vibration sense ; swollen thighs and buttocks; pain on passive stretch	Extreme leg pain ; exclude deep vein thrombosis (DVT) by duplex
Risk factors	Obesity; long operation	Obesity, peripheral vascular disease, major blood loss, muscular calves , intraoperative hypotension
Action	Orthopaedic consult ; compartment pressures; early fasciotomy prevents rhabdomyolysis, creatine kinase surge, kidney injury	Fasciotomy if >30 mmHg and symptomatic; >50 mmHg for >3–4 h → irreversible muscle damage
Prevention	Limit time; padding	No excessive leg elevation; lower legs every 2 h ; avoid ankle dorsiflexion at set-up; pad

Risk Factors and Diagnosis

- **Delayed diagnosis is disastrous**; the whole team should know the risk factors, for prevention and early recognition.

Concept — *positional injury is a product of pressure and time: every preventive measure lowers one (padding, limb angles) or the other (lowering the legs, shorter operations), and obesity worsens both.*

URINARY COMPLICATIONS

- Urine leaks are **common** after upper and lower tract, extirpative and reconstructive surgery, adding morbidity and stay; management is similar open or MIS. **Intraoperative detection and repair reduce morbidity and avoid secondary procedures**, but many leaks appear days to weeks later.

URINE LEAKS AFTER UPPER TRACT UROLOGIC SURGERY

- Follow partial nephrectomy and pyeloplasty; the ureter is also injured in gynaecological and general surgical retroperitoneal operations. Same principles throughout.

Urine Leaks After Partial Nephrectomy

- **Second only to haemorrhage as a complication of partial nephrectomy.**

- **Incidence falling:** historically about 1–17% open and laparoscopic, **below 1% in expert robotic series.** Early low laparoscopic rates reflected selection (large, endophytic, hilar tumours went open). No review shows a significant difference between techniques; robotic surgery may improve ischaemia time, margins and renal recovery over laparoscopy.
- **Risk factors:** tumour size, **endophytic** growth, **warm ischaemia time**, operating time, **hilar location**.
- **Presentation:** after transperitoneal MIS, ileus, fever, abdominal pain; after open surgery, **contained retroperitoneally** (ileus rare). Mean **about day 13 (3–32)**; most are readmitted.
- **Drains:** benefit uncertain; many omit one after MIS because urinomas are rare.
- **Diagnosis:** **computed tomography (CT) urography with delayed images is the mainstay** (extent and site). **Drain creatinine >48 h after surgery** confirms a fistula but not its site or size.
- **Prevention:** find collecting-system entry with **retrograde dilute methylene blue** via a pre-placed ureteric catheter; **close watertight with absorbable suture before renorrhaphy**.

Management

- **Conservative management is the mainstay** after any approach: most leaks heal with drainage and no infection, sometimes needing a drain long term.
- **Trap:** high early drain output after MIS may be **pooled blood and irrigant** redistributing once the patient is supine or mobile, not urine; check drain creatinine.
- **Prolonged leak (>30 days)** is linked to variant pelvic anatomy and age; a minority need secondary intervention; healing can take **80 days or more**.

Escalation: drain creatinine → **keep or reinsert the Foley** (bladder pressure reaches the upper tract) → **ureteric stent with retrograde pyelogram** → **drain off suction and withdrawn in stages**, since a drain against the leak **acts as a siphon** → **interventional radiology (IR) drainage** of undrained or growing collections → second stent, nephrostomy, or ureteroscopic calyceal fulguration with fibrin sealant (desmopressin rarely) → refractory infected leak: **nephrectomy, last resort**.

- **Skin:** urostomy appliance over a leaking drain site; the patient logs output to track resolution.

Concept — *an upper-tract leak heals once the **path of least resistance runs down the ureter**: each step lowers downstream pressure or removes a competing low-pressure exit.*

URETERAL INJURIES

Incidence and Anatomic Landmarks

- **Possibly the commonest surgical complication urologists manage**, often as an intraoperative consult.
- **Hysterectomy:** under 1% despite sealing devices, and **most unrecognised at operation**; laparoscopic hysterectomy injuries lie near the **uterine artery and uterosacral ligament**. Laparoscopic colorectal surgery about 1%.
- **Delayed recognition** → urinoma, ileus, sepsis, shock; **nephrostomy need about tenfold** that of early recognition.
- **Risk factors:** pelvic radiotherapy, malignancy, previous surgery, adhesions, extensive retroperitoneal lymph node dissection (RPLND).
- **Prophylactic ureteric catheters: no robust evidence**; may cause oedema, obstruction and acute kidney injury (AKI).
- **Gynaecological anatomy:** ureter runs with the ovarian vessels, **beneath the uterine artery**, beside the uterosacral ligament; vulnerable at the pelvic inlet and infundibulopelvic ligament.

Mechanisms of Ureteral Injury

- **Spectrum:** pinpoint thermal (usually silent) → ligation → partial or complete transection. **Transections are most often identified** and repaired at once; **ligation and thermal injury present late**; devascularisation → delayed stricture or necrosis.
- Rare in urological laparoscopy. **Identify the ureter early, before ligating any critical pedicle.**
- **Danger sites and landmarks:** gonadal vessels and lower renal pole; pelvic brim, where it crosses the common iliac artery (best place to find and trace it); **under the uterine vessels**; **posterior seminal vesicle dissection** in robotic prostatectomy. Most useful in retroperitoneal fibrosis and complex partial nephrectomy.

Intraoperative Management

INTRAOPERATIVE URETERIC INJURY — MANAGEMENT

Injury	Management
Needle through the full wall; limited thermal injury	Remove the suture; endoscopic stent at the end
Partial tear, minimal tissue loss	Primary closure, 4-0 or 5-0 absorbable , over a stent passed through the defect; stent 4–8 weeks
Complete transection, upper two-thirds (above the pelvic inlet)	Ureteroureterostomy: trim, spatulate, stented anastomosis
Complete transection, lower ureter	Ureteroneocystostomy with psoas hitch (MIS or modified Gibson incision)
Extensive loss	Boari flap

- **Principles:** tension-free; stent $\geq 4-8$ weeks; drain.

Presentation of Unrecognized Ureteral Injuries

- Leak → infection, urinary ascites, sepsis; **rising drain output** is a warning.
- After gynaecological surgery, **ureterovaginal fistula** is about 30% of repairs; **ureteroneocystostomy** is the usual repair (about 70%).
- **Imaging:** CT abdomen and pelvis, intravenous contrast, **delayed images of the whole ureter.**

DELAYED URETERIC INJURY — TIMING

Situation	Action
First line, any time	Endoscopic: retrograde stent, or nephrostomy with antegrade stent, bridging partial transections and small leaks
Found ≤ 7 days, endoscopy inadequate	Early repair (inflammation limited)
Later, or endoscopy failed	Stent, nephrostomy or nephroureteral stent → repair after 6 weeks
Uncontrolled sepsis, persisting collections	Immediate repair regardless of timing

- **Drain every urinoma**, even once the leak stops. Stents stay **4–8 weeks**, matching resolution of inflammation.

Ureterovaginal Fistula

- **Delayed presentation, continuous incontinence.** Intravenous methylene blue with the Foley clamped → blue vaginal pad indicates a fistula.
- May **coexist with vesicovaginal fistula (VVF)** → **double-dye pad test:** oral phenazopyridine + intravesical **methylene blue**, catheter clamped, vaginal pad.

DOUBLE-DYE PAD TEST

Pad stain	Fistula
Orange (dye from the kidney)	Ureterovaginal
Blue (dye from the bladder)	Vesicovaginal
Blue distally, orange proximally	Both

Strictures

- **Endoscopy first** (laser, balloon) for short strictures; **surgery if it fails or the stricture is >2 cm**. Retroperitoneal fibrosis makes repair hard.
- **By level**: short upper → **ureteroureterostomy**; lower → **reimplantation with psoas hitch**; mid → **Boari flap**; long upper → **ileal ureter**; long gaps → robotic **buccal mucosa graft**; right side → **appendix**.
- **Indocyanine green (ICG)** with robotic Firefly imaging checks perfusion of trimmed ends. **Open repair** when adhesions or multiple prior operations preclude MIS.
- **Prevention**: preoperative stents; **lighted stents** to offset absent haptic feedback.

Concept — level and length choose the repair; inflammation chooses the timing. *The lower the defect, the more bladder can be brought up (reimplant → psoas hitch → Boari flap); higher or longer gaps need ureteroureterostomy or interposition. Repair within 7 days or after 6 weeks, with drainage bridging the interval.*

LOWER TRACT URINARY COMPLICATIONS**Bladder Injuries**

- **Rare in urological MIS**; commoner in gynaecological (raising the bladder flap at **hysterectomy**) and colorectal surgery.
- **At operation**: **two-layer absorbable closure**; **catheter 7–10 days** to heal and **prevent VVF**.
- **Postoperatively it mimics ureteric injury** (distension, ileus, fever, rigors, leucocytosis, rising drain output); diagnose by **CT cystography**.
- **Intraperitoneal → two-layer repair** (robotic or open); **small contained extraperitoneal → catheter alone**; extensive post-hysterectomy injury → definitive repair (fistula risk).
- **Commonest urological bladder injury**: posterior bladder-neck “**buttonhole**” in robotic radical prostatectomy (RARP) at the vesicoprostatic junction. Keep the plane without stripping the bladder neck; **identify the ureteric orifices** when difficult (prior transurethral resection, microwave therapy, large median lobe); **5-Fr feeding tubes** or preoperative stents help the less experienced.
- **Bladder cuff leak** (distal ureterectomy, nephroureterectomy) resolves with **7–10 days** of drainage.

Urine Leaks After Radical Prostatectomy**Overview**

- Urethrovesical anastomotic leak is **among the commonest early complications (days 7–10)**, up to about 15%.
- **Definition**: **rising drain output with proportionally falling catheter output**, confirmed by **extravasation on cystography**, usually **posterior** (weakest point).
- Removal on day 2 after posterior reconstruction did not increase leaks, but **most keep the catheter 7–10 days**.
- Graded by **Clavien-Dindo**, in practice by the intervention needed.

- **Significant leaks** collect extraperitoneally (open) or intraperitoneally (robotic) → ileus, longer stay, intravenous fluids; **fever or sepsis** → **antibiotics or percutaneous drainage**. Reintervention about 0.5% open to 2.5% in early MIS series, falling with a **watertight running anastomosis**.
- **Sequelae: bladder neck contracture**; incontinence if scar involves the external sphincter.

Contributing Factors

- **Postoperative bleeding: haematocrit fall >6 points** marks risk. Also **obesity**, operating time, **bladder neck size**, absent bladder neck reconstruction.
- **Barbed versus monofilament suture: no difference**.
- **Salvage prostatectomy**: poor anastomosis in treated tissue; **biograft scaffolding** reduced leaks. **Anterior or total reconstruction** leaks less than conventional apposition.

Diagnosis

- Large leaks are obvious; **raised drain creatinine** confirms. **Retrograde cystography** is valuable; **CT cystography** if fever and leucocytosis suggest a collection.
- Ensure the catheter is in and draining. **Never apply traction**: it disrupts the anastomosis and risks contracture.
- **Healing is time-dependent**: most leaks found on day 8 close by 2–3 weeks. Conservative measures buy time if there is no intraperitoneal leak, ileus or sepsis.

Management

- **Most leaks are small and settle with prolonged catheter drainage; adequate bladder drainage is the key.**

Escalation: confirm catheter patency; change to a **side-fenestrated catheter** (radiological guidance if needed) → drain against the anastomosis on CT: **reposition** (siphon), **off bulb suction** → **IR drainage** of large collections, especially once the drain is out → Foley on **low intermittent wall suction** (vented 18-G angiocatheter) or underwater seal (Pleur-evac); **continuous wall suction is counterproductive** (mucosa blocks holes) → nephroureteral stents on suction or nephrostomy (divert but do not keep the anastomosis patent; rarely needed) → **cystoscopic open-ended ureteric catheters** exteriorised through the urethral catheter (only for partial disruption with easily seen orifices; risks total disruption) → **formal reconstruction** (robotic after RARP).

- **Early formal repair** of major disruption reduces ileus, infection, stay, incontinence and contracture. **After day 7 repair is rarely attempted** (friable tissue): **CT cystogram**, **percutaneous drain**, **parenteral antibiotics** if febrile, maximal drainage; recovery is slow but follows.

UPPER TRACT LEAK VERSUS ANASTOMOTIC LEAK

	Partial nephrectomy	Radical prostatectomy
Confirm	Drain creatinine; CT urography, delayed images	Drain creatinine; retrograde or CT cystography
First step	Keep or reinsert the Foley	Patent, side-fenestrated catheter; no traction
Next	Ureteric stent	Low intermittent suction on the catheter
Drain	Off suction; withdraw or reposition away from the leak (siphon); IR drainage of collections	
Diversion	Second stent, nephrostomy	Nephroureteral stents, nephrostomy, open-ended ureteric catheters
Last resort	Nephrectomy	Formal reconstruction , before day 7

Concept — *anastomotic healing depends on time and drainage: each step improves bladder drainage or diverts urine, and formal repair is reserved for major disruption before day 7, after which tissue will not hold sutures.*

VASCULAR COMPLICATIONS

Overview

- **The most dreaded complications:** among the commonest major MIS complications and the **commonest intraoperative complication**. Audit showed serious complications falling; recent reports do not, reflecting new surgeons and more complex MIS.
- Occur at **access, dissection or vessel isolation**. **Most major injuries are intraoperative; most postoperative bleeding is from unrecognised small vessels**.
- **MIS handicaps:** **no manual compression**; instruments limited by ports; sponges hard to introduce.
- **Early recognition, calm response**. Major vessel injury → **immediate open conversion with manual compression** and emergency vascular consultation.

Intraoperative Vascular Complications

- **2–3% of urological MIS** (non-urological: about 0.5% minor to 20% major, e.g. hepatic resection).
- **Upper tract most prone** (renal vessels off the aorta and IVC; many vessels in partial nephrectomy); lower tract mainly with **extended iliac node dissection**.
- **Haemorrhage is the commonest intraoperative complication** (about 2%; about 40% of perioperative complications), especially near the great vessels (RPLND).
- **Vascular injury is the commonest reason for open conversion**; emergency conversion is more frequent in **laparoscopic live-donor nephrectomy**. Conversion about 1.5%, death about 0.2%.
- **Risk factors:** previous surgery, concomitant abdominal disease, obesity, aberrant anatomy, **inexperience** → careful early case selection.

Management

- **Cognitive reaction first:** control your stress response, alert the team to a possible **massive haemorrhage protocol**; after temporary control, **pause and state the contingency plan**.
- **Benchmark:** react about 22 s, control <1 min, repair about 3 min (about 170 s); longer if only temporarily controlled.

Consensus algorithm (Passerotti): raise pneumoperitoneum **cautiously** (**gas embolus** risk with venous injury) → aspirate, expose → working suction, **laparoscopic sponges** → add ports promptly, away from other trocars → identify every injury → clips **judiciously, never randomly** (failed clips obstruct suturing and stapling) → **suture repair, the most reliable rescue** → robotic: assistant presses with the suction tip; selective vascular clamps → **low threshold for conversion and vascular consult**.

Vascular Injury During Abdominal Access

- **About 0.1%** (about 0.04% with blind insufflation); Veress needle or trocar.
- **Blood aspirated through the Veress needle** → **stop insufflating** (CO₂ embolism).
- **Gas embolism:** sudden hypoxia, hypercarbia, hypotension, arrhythmia, head and upper-limb cyanosis, “mill-wheel” murmur, **falling end-tidal CO₂**, raised right heart pressure. Gas in the right heart or pulmonary artery → reduced venous return and cardiac output → pulmonary hypertension → right heart failure. **Mortality is high once blood pressure or heart rate falls**; transoesophageal echocardiography detects 0.02 mL.

Treatment: desufflate → 100% oxygen (FiO₂) → Trendelenburg and left lateral decubitus (trap gas in the right atrium) → central line aspiration.

- **Leave the offending needle or trocar in place;** alternative access locates the injury, and the trocar may tamponade it.
- **Veress versus trocar:** a Veress puncture of the aorta or inferior vena cava (IVC) may self-limit if the haematoma is not expanding (still inspect directly); trocar injuries **almost always need repair, usually with conversion; through-and-through** injuries are reliably repaired only open. Multidisciplinary: resuscitation, vascular surgery; experienced surgeons may repair by MIS.
- **Pressures:** 15–18 mmHg for trocar insertion → 12–15 mmHg thereafter.
- **AirSeal:** valveless trocar; stable pneumoperitoneum; smoke evacuation; **less wall and diaphragm tension** (easier ventilation in steep Trendelenburg or flank). The laxer wall means **all trocars go in under standard insufflation before AirSeal mode.**

Mechanisms of Injury

- **Umbilicus** lies from 5 cm cephalad to 3 cm caudal of the iliac bifurcation, so it is **unreliable**; the **anterior superior iliac spine (ASIS)** has been proposed instead.
- **Veress angle:** 45° at normal body mass index (BMI), 90° in obesity, as the umbilicus moves caudally with rising BMI.
- **Vessel most often injured at the umbilicus:** left common iliac vein.
- **Skin to retroperitoneum** about 6 cm, as little as 2 cm when thin; Trendelenburg shortens it.
- **Atherosclerotic aorta or abdominal aortic aneurysm (AAA):** rigid wall, **no spontaneous tamponade**, “mini-fracture”, plaque embolism to the legs → **enter at the left upper quadrant**; review or obtain imaging.
- **Flank position:** upper quadrants are classic safe sites (right: **liver** over upper pole; left: **spleen** only if enlarged). Robotic kidney: Veress at the future camera port (**two fingerbreadths above the umbilicus, lateral rectus line**), **perpendicular to the posterior peritoneum** (bowel falls medially). Extensive previous surgery → **Hasson or optical trocar.**

Comparison of Different Access Techniques

- Under-reporting hampers comparison. In U.S. Food and Drug Administration (FDA) data, **81% of trocar deaths were from major vessel injury (aorta, IVC, iliacs; iliac rarely fatal).** **No consensus** on the safest technique.

ACCESS TECHNIQUES COMPARED

Technique	Strengths	Limitations / evidence
Veress (closed)	Most widely used	Old reports: slightly more vascular injury; Cochrane: only more access failure
Open Hasson	Classically safest for novices and adhesions	Slowest to establish pneumoperitoneum
Optical bladeless	Visual feedback through layers; fast; smaller fascial defect	Rare, mostly minor injuries (mesenteric, epigastric)
Axial-force-limiting (sheath, then conical port)	Pushes vessels aside; countertraction limits force	Injuries arise mainly from excess forward force

- **Trocar technique:** rotate; **two hands** (dominant on top, non-dominant on the shaft at skin level to **prevent plunging**); **skin incision wide enough for the shaft.**

Management of Vascular Incidents

- Suction, irrigation, direct vision; open control above and below; **extra suction**; vascular help as needed.
- **Transient pneumoperitoneum 20–25 mmHg** (embolism caution) defines and temporises even medium–large venous defects, **for no more than about 10 minutes** (CO₂ absorption, splanchnic hypoperfusion).
- **Pressure** (gauze or rolled Surgicel on the assistant's suction) → accessory trocars → continue by MIS if controlled → **convert if the injury cannot be defined or controlled; transfuse early**, before instability.

Injury to Epigastric Vessels

- **0.2–2%; often presents late** because the trocar tamponades it.
- Vessels run in the **lateral rectus sheath**; injured when a trocar angles **medially**. **Prevent: transilluminate; lateral ports ≥5.5–6 cm from the midline; inspect sites as trocars come out.**
- **Repair:** Foley balloon traction temporises (rarely needed); **best is a fascial closure device (Carter-Thomason)** placing a figure-of-eight under vision without enlarging the incision.

Injury to Major Vessels

- **The most feared** (taught by simulation). Most **cannot be ligated (no collaterals)** → repair, usually after conversion.
- Suspected trocar injury: **leave it in place** (tamponade), add ports if stable, convert through a **midline incision**.
- **Robotic rescue:** ProGrasp temporises; other arms and suction assess; **rescue stitch** of polyglactin on a **CT-1 needle with a Hem-o-lok at the tail** tents the wall between throws; the large needle is findable in blood.
- **Ligatable: internal iliac and inferior mesenteric** (good collaterals).
- **Superior mesenteric artery (SMA):** in the lateral position the left renal vein no longer marks the SMA–aorta relation; the SMA is taken for the left renal artery when tumour distorts anatomy or retraction is poor. **Suspect it if the artery is medial to the aorta, anterior to the renal vein, or atypical** → mobilise bowel, medialise mesentery, **expose the aorta if in doubt**. Variant or multiple renal arteries (sometimes missed on imaging) occur in over half of patients.
- **SMA stump or proximal trunk injury: mortality up to 75%.** Backflow via the **marginal artery of Drummond** and **arc of Riolo** may help; **no distal flow → repair** (re-anastomosis, splenic or left renal stump transposition, graft). **Vascular consult and conversion are mandatory.**
- **Retroperitoneal haematoma: stable, non-expanding → observe, do not explore**, continue or abandon by urgency; **unstable or expanding → open repair** (aorta, IVC, external iliac artery).
- **Venous tears:** insufflate to **20 mmHg**, suture quickly, return to **12–15 mmHg** promptly. **External iliac vein:** also **turn off sequential compression devices** to decompress it. The robotic **fourth arm** gives clamping and tension; use sealing devices near major structures.

Bleeding During Tumorectomy in Partial Nephrectomy

- Robotic partial nephrectomy (RAPN) is the **commonest approach**; routine artery-and-vein clamping is declining. **Right-sided tumours bleed more.**
- **Artery-only clamping:** more blood loss than full hilar clamping, **no more transfusions**. Add a venous clamp if bleeding halts progress; **parenchymal venous backflow**, by contrast, slows towards the crater.

RIGHT VERSUS LEFT RENAL VEIN IN PARTIAL NEPHRECTOMY

Right	Left
Short, often ≤2 cm ; open IVC stump gives torrential backflow	Long (up to 10 cm); “mesenteric claw” and pneumoperitoneum keep backflow minimal

Right	Left
Clamp the vein, especially if ≤ 2 cm	Clamp only for complex tumours (hilar, endophytic, perihilar, hypervascular), distal to gonadal and adrenal veins

- **CT angiography or magnetic resonance angiography (MRA)** is required for selective or superselective clamping and to find **accessory arteries**.
- **Torrential bleeding despite clamping** (missed artery, or venous clamp proximal to a lumbar vein) → **unclamp the vein** to restore venous return.
- **Imaging matched with intraoperative ultrasound** determines safety.

Stapler Malfunction

- **About 1%** accepted. FDA reviews: deaths from misfiring and failed or malformed staple lines; recalls for broken parts or sterility; **under-reported** (most programme directors had seen one; a quarter changed the operation).
- Commonest urological use: **renal hilum**. **Commonest cause: tissue too thick**.

Prevention: inspect on unpacking → load precisely → dissect the vessel → **see the whole jaw and tip before firing** → without tumour thrombus, **fire close to the kidney**, leaving room for a “rescue” stapler → **no clips in the jaws** (jamming) → **no excess handle force** (overrides the lockout).

HILAR CONTROL DEVICES — FAILURE MODES

Device	Leading failure
Stapler	Staple-line malformation; lockup . Rescue: extra clips over the staple line or below the transection
Locking clip (Hem-o-lok)	Dislodgement
Titanium clip	Closing or cutting failure; applier jamming

- Most prefer staplers (no dislodgement), but **FDA data show no device safer**.

Postoperative Vascular Complications

ARTERIAL VERSUS VENOUS POSTOPERATIVE BLEEDING

Arterial	Venous
Rapid haematocrit fall ; instability	Smaller vessel; self-limiting ; presents later
Ligation → organ infarction , delayed presentation	Pneumoperitoneum tamponades it → delayed bleed

- Usually **unrecognised intraoperative injury**. **Vascular complications are most early postoperative complications after laparoscopy**; injury can occur **along instrument shafts** or at entry and exit, outside the field of view.
- **Partial nephrectomy: most haemorrhages occur after discharge**. **Arteriovenous (AV) fistula and pseudoaneurysm** (commoner) appear at **2–3 weeks**, in about 2–3%, roughly twice as often after MIS as open.
- **Mechanism:** adjacent segmental artery and vein injured in the bed or by suture; risk rises with **depth and nephrometry score**; early unclamping may reduce it.
- **Prevention:** inspect the bed; **ligate larger segmental branches**; running suture through the crater; **avoid deep renorrhaphy sutures**.

Diagnosis and Management

- **Precipitous haematocrit fall** is the usual clue; **haematuria** adds alarm. Up to **10%** are **asymptomatic**; routine early CT finds some, nearly all silent.

- **<2–4 mm pseudoaneurysms resolved spontaneously.** May mimic recurrence (arterial-phase enhancement); equivocal CT → **angiography**.
- **Presentation:** fever, gross haematuria, flank pain, hypotension, bloody drain discharge, dizziness, syncope; a minority need **emergency embolisation**, especially with **multiple pseudoaneurysms**.
- **High suspicion after recent partial nephrectomy → direct angiography, bypassing CT:** spares an anaemic, depleted, renally impaired patient extra contrast, and is faster.
- **Incidental, with no haematuria, haematocrit fall or instability → conservative,** with serial haematocrit and observations.
- **Delayed haematocrit fall → CT (arterial-phase blush);** unstable or steep fall → angiography.
- **Selective angioembolisation is the treatment of choice** for pseudoaneurysm and AV fistula: **superselective microcoils** succeed technically in nearly 100%, with some parenchymal loss but **no harm to estimated glomerular filtration rate (eGFR)**. Failure → surgery.

Concept — intraoperative bleeding is controlled first by pressure and a pause, then repaired by suture; clips and staplers can fail and obstruct the rescue. Bleeding after partial nephrectomy is a **delayed arterial lesion** that angiography diagnoses and treats in one step.

BOWEL INJURIES

Incidence and Presentation

- **Under-reported** (minor serosal tears); **about 0.2%.** Missed injury → ileus, pain, leucocytosis, sepsis, significant mortality.
- **Up to about two-thirds are unrecognised at operation,** occurring anywhere from entry to exit. Incidence has risen slightly with complex MIS, but the **unrecognised proportion has fallen markedly.**

MISSED BOWEL INJURY — CLASSIC VERSUS POST-LAPAROSCOPIC (BISHOFF)

Classic (open)	After laparoscopy
Peritonitis, acute abdomen	Often absent
Leucocytosis	Leucopenia
Generalised pain	Single trocar-site pain out of proportion; no purulence
—	Onset 24–48 h; distension, ileus, diarrhoea

- **CT with oral and intravenous contrast, delayed images; rectal contrast** if rectal injury suspected. **Free gas is normal for up to 7 days** after laparoscopy. Some still present classically.

Mechanisms of Injury

- **Causes:** trocar technique, Veress needle, over-retraction, handling, **electrocautery**. About a fifth of trocar access injuries in FDA data were bowel injuries, some fatal.
- **Midline scar → Veress away from it;** adhesions may be remote, so use an **optical trocar or Hasson**.
- **Veress injuries are mostly small bowel. Saline drop test;** after partial insufflation the needle advances **1–2 cm freely** (usually intraperitoneal).
- After the camera port, **inspect for adhesions** at planned port sites; **insufflate fully** (no peritoneal tenting); **bowel preparation** for difficult or multiply operated cases.
- **Adhesiolysis with cold scissors** (no thermal spread); **countertraction** against shear and denuding injury, especially robotically (**no haptic feedback**, strong ProGrasp grip). Only the adhesiolysis the field needs; **no “sweeping”**.

- **Mechanical injury** (abrasion, compression, sharp) when instruments move blindly beyond their safe range; look for **multiple injuries** (they decide repair versus resection); open the working space widely; **convert** if adhesions prohibit safe MIS.

BOWEL INJURY — MANAGEMENT

Injury	Management
Small bowel perforation or serosal tear	Primary suture repair
Limited colonic, no major spillage	Primary repair (open or MIS)
Extensive laceration; mesenteric injury, doubtful viability	Resection, primary stapled anastomosis ; general surgery consult
Extensively diseased bowel	End ileostomy , later reconstitution
Extensive colonic with faecal soiling	Colostomy ; irrigation; broad-spectrum antibiotics
Superficial thermal burn	Serosal Lembert sutures; never conservative
Extensive burn	Debride and close ; damage exceeds visible discolouration → resection ± conversion

- **Most delayed injuries are thermal. Monopolar:** critical spread >20 mm (2 s at 60 W). **Bipolar is the modality of choice.** Others spread a few millimetres: **harmonic < bipolar < LigaSure < robotic hot shears.**
- **Stray current:** poor technique or **insulation defect**; **direct or capacitive coupling** with metal trocars.
- **Prevention:** sparing monopolar (still used on robotic scissors); prefer harmonic or vessel sealers; bowel away from the tip; **energised tip always in view**; inspect shafts beforehand; **inspect the field on exit.** Robotic: console surgeon watches insertions; assistant inserts slowly under vision until instrument memory is set and reports resistance.

Duodenal and Pancreatic Injuries During Nephrectomy

- **Duodenum: right renal surgery (Kocherisation).** Shear or puncture; **even <5 mm injuries perforate** → delayed fever, pain, sepsis. **Intramural haematoma** → prolonged obstructive ileus, bowel rest, TPN.
- **Pancreatic tail: left nephrectomy,** near the upper pole and adrenal → **fistula, pseudocyst, abscess.**
- **Octreotide** reduces morbidity after pancreatic resection and in severe pancreatitis, but **empirical use after injury is questioned** (more fistulae, longer drainage and stay).
- **Obvious injury → distal pancreatectomy.** Delayed: clear drain fluid with **high amylase** → CT for collections → extra drains; no drain → percutaneous drainage, somatostatin, bowel rest, nasogastric suction, nutrition; persistent → distal pancreatectomy.

Rectal Injury

- **Commonest bowel complication of radical prostatectomy;** a learning-curve concern.
- **Laparoscopic highest (about 1%);** open and robotic similar (under 0.5%); **salvage about 7%.** Risk factors: salvage, stage, Gleason grade, nodal status, experience, **not approach.**
- **Recto-anastomotic fistula under 0.1%;** predictors: laceration extent, prior radiotherapy, rectal infiltration.
- Also difficult after TURP, microwave thermotherapy, recurrent prostatitis (fibrosis).
- **Commonest RARP site: near the apex in posterior dissection. No cautery on the anterior rectal wall.** Rectal EEA dilator identifies the rectum in fibrotic cases; **modified bowel preparation** before salvage surgery.

Management

- **No gross spillage → primary closure:** debrided, vascularised, **tension-free**; multilayer non-overlapping; mucosa and serosa absorbable, perirectal bolster non-absorbable; usually two layers.

- **Bubble test** (air via **Toomey syringe** or rectal tube); **omental or fat interposition** if possible; **drain mandatory**; **low-residue diet** and **broad-spectrum antibiotics 3–7 days**.
- **Minor**, prepared rectum → **primary repair**; **moderate–extensive** after **salvage (irradiated) surgery** → **diverting colostomy**.
- **Unrecognised**: early **leucopenia (MIS)** or **leucocytosis (open)**, nausea, vomiting, ileus, **lateralising pelvic pain**, peritonitis → **CT with rectal contrast** or **Gastrografin enema** → **exploration, irrigation, diverting colostomy** (prevents sepsis and **rectourethral fistula**).

Port Site Hernia

- **0.3–5.4%**; risk: older age, previous hernia, operating time, **trocar size**.
- **Painless bulge**, mostly at **extraumbilical ports**, within weeks (mean about 10 days); CT confirms.
- **12-mm ports** far exceed **10-mm**; use **5-mm** where possible. **Bladeless trocars** leave **smaller defects**, though 12-mm bladeless hernias occur. Inspect **robotic 8-mm off-midline ports** (enlarged by arm torque).
- **Single-site surgery** (SILS, single-incision laparoscopic surgery; LESS, laparoendoscopic single-site surgery): better cosmesis, **more incisional and umbilical hernias** (larger incision); about 5% with da Vinci Xi single-site, better with **da Vinci SP** (smaller incision, semi-flexible instruments); more data needed.
- **Close large fascial defects**: single **figure-of-eight absorbable** suture, usually with a closure device. Symptomatic hernia → **laparoscopic or robotic mesh repair**.

Venous Thromboembolism

- **Venous thromboembolism (VTE) risk**: pelvic surgery, malignancy, obesity, long operations (MIS is longer on the learning curve). Pneumoperitoneum and reverse Trendelenburg might cause stasis, but there is **no evidence of more DVT after laparoscopy**; true DVT and pulmonary embolism (PE) incidence is unknown.
- **MIS prostatectomy**: symptomatic VTE about 0.5%, mostly DVT alone, occasional fatal PE; associated with smoking, large prostate, re-exploration, long operation and stay.
- **By procedure** (American College of Surgeons National Surgical Quality Improvement Program, ACS NSQIP): about 1% for nephrectomy, partial nephrectomy, nephroureterectomy and prostatectomy; lower for MIS nephrectomy; **about 5% for radical cystectomy**. **Most VTEs occur after discharge**.
- **Preventer trial** (Brady Institute, randomised): perioperative subcutaneous heparin after prostatectomy **did not reduce symptomatic VTE**.
- **Cystectomy**: about half of VTEs after discharge; **discharge on low-molecular-weight heparin (LMWH; enoxaparin)** roughly halved VTE; **oral anticoagulants** appear similarly safe and effective.
- **No consensus guidelines**; practice varies (about a third of prostatectomy patients get none) → **follow local policy**.
- **American Urological Association (AUA) best practice statement 2009**: **pharmacological prophylaxis mandatory at high risk** (age >60, prior VTE, cancer, hypercoagulable state).
- **Risk factors**: age, personal or family VTE, malignancy, obesity, thrombophilia, and **transfusion** (after cystectomy, each unit raises VTE risk by about 7%); **restrictive transfusion lowers VTE** without raising mortality or cardiac events.
- **Caprini score** (modified by the American College of Chest Physicians, ACCP): most widely used, **validated in urology**; summed weighted points: **very low 0**; **low 1–2**; **moderate 3–4**; **high ≥5**.

TABLE 20.2 THE CAPRINI RISK ASSESSMENT MODEL (GROUPED)

Points	Age	Surgery / immobility	Thrombotic history	Other
1	41–60 y	Minor surgery; medical patient at bed rest	—	BMI >25; swollen legs; varicose veins; pregnancy or postpartum; unexplained or recurrent spontaneous abortion; oral contraceptive or HRT; sepsis, serious lung disease or pneumonia (<1 mo); abnormal pulmonary function; acute MI; heart failure; inflammatory bowel disease
2	61–74 y	Arthroscopic; major open >45 min; laparoscopic >45 min ; bed >72 h; plaster cast; central venous access	—	Malignancy
3	≥75 y	—	Personal or family history of VTE ; factor V Leiden; prothrombin 20210A; lupus anticoagulant; anticardiolipin antibodies; raised homocysteine; heparin-induced thrombocytopenia; other thrombophilia	—
5	—	Elective arthroplasty; hip, pelvis or leg fracture	—	Stroke or acute spinal cord injury (<1 mo)

From Gould MK, Garcia DA, Wren SM, et al.: ACCP prevention of VTE in nonorthopedic surgical patients, 9th ed. Chest 2012.

- **Pattern:** 1 point = common medical conditions; 2 = surgery, immobility, cancer; 3 = thrombotic history and thrombophilia; 5 = major orthopaedic or neurological events.
- **Traps:** **laparoscopic >45 min scores the same as open**; malignancy is 2, not 3; **family history equals personal history (3)**. A 61–74-year-old with cancer having major surgery is already **high risk**.

TABLE 20.3 RECOMMENDATIONS FOR THROMBOPROPHYLAXIS IN VARIOUS RISK GROUPS

Risk of symptomatic VTE	Bleeding risk average (~1%)	Bleeding risk high (~2%) or severe consequences
Very low (<0.5%)	No specific prophylaxis	No specific prophylaxis
Low (~1.5%)	Intermittent pneumatic compression (IPC)	IPC
Moderate (~3.0%)	Low-dose unfractionated heparin (LDUH), LMWH or IPC	IPC
High (~6.0%)	LDUH or LMWH <i>plus</i> IPC	IPC until bleeding risk falls, then add pharmacological
High-risk cancer surgery	LDUH or LMWH <i>plus</i> IPC, and extended LMWH after discharge	As above
High risk; LDUH and LMWH contraindicated or unavailable	Fondaparinux or low-dose aspirin (160 mg) ; IPC; or both	As above

From Gould MK et al.: ACCP 9th ed. Chest 2012.

- **Pattern:** rows are thrombotic risk, columns bleeding risk; **high bleeding risk means IPC until it falls**; only cancer surgery earns **extended LMWH**.

- **Traps:** very low risk needs only ambulation; aspirin and fondaparinux are for when heparins are contraindicated.
- **By tier:** very low (most transurethral and anti-incontinence surgery) → **early ambulation** ± stockings; low → **IPC or graduated stockings**; moderate → **pharmacological, preferably LMWH** (LDUH or fondaparinux acceptable); high or very high → LMWH, LDUH or fondaparinux **plus mechanical**. **Major open surgery** warrants a more aggressive combined regimen than MIS.
- **Timing:** **mechanical just before surgery**; pharmacological **2–12 h preoperatively**; **fondaparinux 6–8 h postoperatively**. Subcutaneous heparin can start just before or during most urological procedures; starting **within 48 h** did not increase bleeding.
- **Duration:** until **fully ambulatory** (poorly defined); risk **persists for weeks**. Extended prophylaxis is not routine **except after major abdominopelvic cancer surgery**: **LMWH ≥1 month** markedly cut total and symptomatic VTE without extra bleeding (meta-analysis of randomised trials).
- **Anticoagulation contraindicated or high bleeding risk** → mechanical before surgery; add pharmacological once haemostasis is secure.

Concept — *VTE prophylaxis is a two-axis decision: thrombotic risk (Caprini) sets intensity, bleeding risk chooses mechanical or pharmacological, and duration follows the fact that risk outlasts the admission, above all after cystectomy and cancer surgery.*

Lymphoceles

- **The commonest complication of lymphatic disruption after pelvic lymph node dissection (PLND):** lymph collections **without epithelial lining**, usually along the **iliac vessels**. First described about 50 years ago (Basinger and Gittes).
- **Transperitoneal RARP** was expected to reduce them (peritoneal reabsorption), but **loculation and retroperitonealisation occur within 48–72 h**; in the robotic era, **approach makes no difference** to symptomatic lymphoceles.
- Rates vary with template (most are silent); **extended dissection roughly doubles them**.
- **Lymphostasis:** clips versus bipolar (no robust data; some favour clips); **robotic vessel sealer**; **fibrin sealants**; **peritoneal flap** fixed to the bladder. **Inspect and control visible lymphatics** after PLND or radical upper tract surgery; **drain** most major nodal dissections.
- **Management:** **asymptomatic and <5 cm → conservative**; **large symptomatic** (iliac compression → leg oedema; bladder compression → frequency, urgency) → **percutaneous drainage**; **thick-walled or recurrent after drainage → laparoscopic marsupialisation into a peritoneal window**. **Aspiration is not advised** (always reaccumulates; infection). **Sclerotherapy** (povidone-iodine, doxycycline) may hinder later marsupialisation. **Lymphangiography with embolisation:** high success, faster fall in output than sclerotherapy, occasional reintervention; long-term data awaited.
- **Infected lymphocele:** fever, rigors, sepsis → intravenous fluids, broad-spectrum antibiotics, percutaneous drainage; **always consider a urine leak**.

Concept — *lymphoceles are treated by symptoms and recurrence, not size alone; the definitive cure is internal drainage into the peritoneum, which is why aspiration and sclerotherapy, which compromise it, are avoided.*

SECOND VICTIM SYNDROME

- **The surgeon's own distress after a serious complication.** In surgery responsibility rests on **“the hands of the surgeon”** rather than diffusing through the system. **Almost half** of healthcare providers experience it.
- **Risk factors:** lapse in judgement or concentration, inexperience, burnout, personal problems.

- **Course: adrenaline surge** → failure, sadness, depression, isolation, unworthiness, **medicolegal fear**; **cognitive impairment for months** (concentration, decisions, dexterity). The detachment that lets surgeons work in chaos breaks down.
- **Culture** is harsh; training covers managing complications, not **supporting peers**.
- **Recovery: confidential forums beyond morbidity and mortality meetings**, trusted seniors, **mindfulness and meditation**. Only a minority (experienced, resilient) report a positive career effect. “The only surgeon without complications is one who does not operate” ignores the influence of the relationship with the first victim.

Concept — *a complication has two victims; the surgeon's impaired cognition afterwards is itself a safety risk, so peer support belongs to complication management.*

CONCLUSION

KEY POINTS

- **Physiological, urinary, vascular, bowel and visceral** complications occur with every form of urological surgery.
- Laparoscopic and robotic surgery bring a **distinct set of complications** and demand new operative, diagnostic and therapeutic skills.
- More complex MIS operations mean **complication rates can rise**.
- MIS complications can present **atypically and late**.
- A **high index of suspicion** enables prompt diagnosis. Many complications are managed conservatively or medically; **diagnostic and interventional radiology** is central; some need operation.
- **Prevention matters most**: patient selection, positioning, planning, meticulous technique.